Yassine Jallouli

AI Software & Data Engineer — Full-Stack Developer
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Summary

Software Engineer focused on Full Stack, AI, and Data Engineering, with hands-on experience building scalable applications, backend systems, and data-driven solutions. Skilled in Angular, Spring Boot, Python, SQL, and LLM technologies including RAG, LangChain, and LangGraph. Experienced with REST APIs, Docker, CI/CD, and Microsoft Azure, with a strong interest in intelligent software, data pipelines, and production-ready AI systems.

Professional Experience

Software Development Instructor *Mar 2026 – Present*

9antra.tn – The Bridge

- Deliver project-based training in Spring Boot and Angular, covering REST APIs, application architecture, Git, debugging, and deployment.
- Mentor learners through the development and deployment of complete full-stack applications.

Final Year Internship *Feb 2025 – Aug 2025*

BeeCoders

- Engineered **ArenaLink**, a full-stack football management platform using Spring Boot, Angular, and MySQL.
- Delivered core features including authentication, payments, real-time chat, live match management, and stadium booking.
- Integrated LLM APIs to enable AI-powered content generation and automated platform features.
- Designed scalable RESTful APIs following clean architecture and role-based access principles.
- Containerized the application with Docker and automated CI/CD workflows using GitHub Actions.
- Deployed the platform to Microsoft Azure, streamlining the delivery process and reducing manual deployment steps.

Software Development Intern *Jul 2024 – Sep 2024*

Expertify

- Developed a full-stack e-commerce platform using Angular and Node.js (Express).
- Built and integrated core features for product browsing, navigation, and checkout workflows.

Software Development Intern *Jul 2023 – Sep 2023*

Expertify

- Built a full-stack web application using Angular and Spring Boot.
- Created reusable frontend components and integrated backend REST APIs.

Software Development Intern *Jul 2021 – Aug 2021*

Formalab

- Developed a scheduling web application using Angular and Node.js.
- Enhanced usability through interactive frontend features and interface improvements.

Projects

London Transport Data Analysis *SQL, Snowflake*

- Analyzed 12+ years of Transport for London journey data (2010–2022) using SQL and Snowflake.
- Identified transportation trends, peak usage periods, and unusual Underground traffic patterns.
- Applied aggregation, filtering, and exploratory data analysis techniques on a cloud data warehouse.

AI Resume Analyzer & Job Recommendation System *Python, LangChain, LCEL, RAG*

- Built an LLM-powered application that extracts structured information from PDF resumes using LangChain and Pydantic.
- Implemented RAG-based job recommendations using Hugging Face embeddings and ChromaDB vector search.
- Generated ATS feedback, resume improvement suggestions, personalized interview questions, and tailored cover letters.

Student Productivity Prediction *Python, Scikit-learn*

- Built a machine learning pipeline using 20,000+ student records.
- Performed data preprocessing, feature engineering, and exploratory data analysis.
- Trained and compared Random Forest, Decision Tree, and Linear Regression models using MAE and R^2.

CI/CD Automation Pipeline *Jenkins, Docker, SonarQube, Prometheus, Grafana*

- Designed an automated CI/CD pipeline covering build, test, quality analysis, and deployment workflows.
- Integrated SonarQube for continuous code quality analysis and Docker for containerized delivery.
- Implemented application monitoring with Prometheus and Grafana.

Education

Engineering Degree in Computer Engineering 2020–2025

Ecole Supérieure Privée d'Ingénierie et de Technologies (ESPRIT)

Baccalauréat – Technical Sciences 2018–2019

Lycée El Wafa

Skills

AI & Data Engineering: Python, SQL, Pandas, NumPy, Scikit-learn, LLMs, RAG, LangChain, LangGraph, Prompt Engineering, ETL, Data Pipelines

Software Engineering: Java, Spring Boot, Angular, React, Node.js, NestJS, JavaScript, TypeScript, REST APIs

Databases & Cloud: MySQL, PostgreSQL, MongoDB, ChromaDB, Snowflake, Microsoft Azure, Docker

DevOps & Tools: Git, GitHub, GitHub Actions, Jenkins, JIRA, SonarQube, Prometheus, Grafana, LangSmith

Certifications

- Associate Data Engineer in SQL – DataCamp (2026)
- Designing Agentic Systems with LangChain – DataCamp (2026)
- Retrieval Augmented Generation (RAG) with LangChain – DataCamp (2026)
- Developing LLM Applications with LangChain – DataCamp (2026)
- Machine Learning Specialization – DeepLearning.AI (Coursera, 2026)
- Hackathon Participation – Sustainable Minds (2023)

Languages

Arabic (Native) — English (Professional) — French (Professional)